

Think Integrals

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1 Introduction

Concern 1: Integrals have been an integral part of mathematics. Mathematicians work on a problem for hours through months, and finally end up giving a solutions.

Concern 2: The best and most desirable part is the problem and solution, but intrinsically the feeble belief of thoughts and approaches that a mathematician works and go through, very few people are concerned about it.

Concern 3: "Think Integrals" doesn't only mean solving integrals, but every problem that I feel to work on deeply, and counter-intuitively on paper. (Intuition rests on the mind, which even a linguist can't explain properly)

Concern 4: "Think Integrals" is where I am supposed to think about the problems, and integrate (or accumulate) all the result and equations to solve the problems at the end.

The commencement of this enactment can be done well with an integral itself. A common problem (maybe!) of integration is

$$P = \int_{x_1}^{x_2} \frac{\ln(x^2+1)}{x^2+1} dx$$

By observing the problem, for the first time I can think of a substitution of $(x^2 + 1)$ by 'u' (as always).

$$u = x^2 + 1 \Rightarrow \frac{du}{2} = dx \quad (1)$$

Before going forth, we can administer something analogous to IBP (integration by parts). Since,

$$d(\ln u) = \frac{du}{u} \quad (2)$$

(And of course, from the initial property of derivatives.)

Now, substituting for u in the expression, we get:

$$P = \int_{x_1}^{x_2} \frac{\ln u}{2u} du \quad (3)$$

Here, I substituted (1) into P.

$$P = \int_{x_1}^{x_2} \frac{\ln u}{2} d(\ln u) \quad (4)$$

From (2).

$$P = \frac{1}{4} ((\ln(u))^2) \Big|_{x_1}^{x_2} = \frac{((\ln(x_2^2 + 1))^2 - \ln((x_1^2 + 1))^2)}{4} \quad (5)$$

Again from (1) Or, to make it even simpler (maybe!) , we have

$$P = \frac{\ln(\frac{x_2^2+1}{x_1^2+1})\ln((x_2^2+1)(x_1^2+1))}{4} \quad (6)$$

Here, I conclude this problem. It was an ordinary yet conceptual problem. It incorporates many concepts such as substitution, analogy to IBP (actually a type of substitution), and of course integral result in (5). Although it isn't appropriate to rest on limits x_1 and x_2 in (3) and (4), since x is no longer the variable, which will be corrected in along way problems.